LONGEVITY 08

Elamipretide (SS-31)

Synthetic 4-amino-acid peptide that binds cardiolipin in mitochondria; injected under the skin



CLASS	STUDIED FOR	HUMAN EVIDENCE	US STATUS
Peptide	Barth syndrome, muscle, eye disease	Controlled trials; FDA approved	FDA approved (Forzinity) · accelerated

STUDIED FOR

- **Barth syndrome** — HUMAN
- **Mitochondrial myopathy** — HUMAN
- **Dry macular disease** — HUMAN
- **Heart attack** — HUMAN
- **Muscle energy, aging** — HUMAN

Approved for Barth syndrome only. Use for aging is off-label. No trial has tested elamipretide for longevity; a small pilot in adults 65 to 80 is recruiting.

HOW IT WORKS IN THE BODY

Elamipretide ▶ Binds cardiolipin ▶ Inner membrane ▶ Mitochondria

- **Mitochondrial cardiolipin binder** that settles in the inner membrane (label)
- **Improves mitochondrial** shape and function, per the label
- **In aging,** an effect in people is not established

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Approval rests on a 12-person trial and extension
- Larger trials mostly missed main goals
- Confirmatory Barth trial running

NOT YET ESTABLISHED

- Effect on aging in people
- Half-life (not on the label)
- Cancer risk (not studied)

QUICK PRACTICAL INFO

HALF-LIFE	ROUTES STUDIED
Not established (not given on the label)	Under the skin (label); IV in some trials
PHARMACOKINETICS	**SOLD AS**
Peak in 0.5 to 1 hour; broken down to inactive peptides; urine	Prescription injection (Forzinity)
MONITORED IN TRIALS	**SPORT (WADA)**
Allergic reactions, injection sites, eosinophil counts	Not on the 2026 list

REGULATORY

FDA	Accelerated approval, Sep 2025, Barth syndrome
FDA	Confirmatory trial required
WADA	Not listed

REFERENCES

1. FDA, Forzinity (elamipretide) prescribing information, Dec 2025
2. FDA, accelerated approval of first treatment for Barth syndrome, Sep 2025
3. WADA, 2026 Prohibited List

Adonis Men's Health · checked Nov 2026. Research and regulator records only. Not medical advice.

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